DEVELOPMENT OF A LOCALIZED REAL-TIME FREQUENCY MONITORING SYSTEM FOR STRUCTURAL HEALTH MONITORING USING MEMS IMUS AND ESP32

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APPROVAL SHEET

This thesis titled **"Development of a Localized Real-Time Frequency Monitoring System for Structural Health Monitoring using MEMS IMUs and ESP32"**, prepared and submitted by **JAN CLYDE T. TALOSIG** in partial fulfilment of the requirements for the degree, Bachelor of Science in Electronics Engineering, is hereby recommended for final defense.

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DEDICATION

This section is a personal note where the researcher dedicates the work to individuals or groups who provided inspiration, support, or motivation. It is usually written in a heartfelt and informal tone compared to the rest of the manuscript.

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ACKNOWLEDGEMENT

This section expresses gratitude to those who contributed to the completion of the study. It can include advisers, faculty members, classmates, family, friends, and institutions. The tone is formal but appreciative.

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ABSTRACT

TITLE : DEVELOPMENT OF A LOCALIZED REAL-TIME FREQUENCY MONITORING SYSTEM FOR STRUCTURAL HEALTH MONITORING USING MEMS IMUS AND ESP32

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INSTITUTION : CAGAYAN STATE UNIVERSITY CARIG CAMPUS

DEGREE : BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING

YEAR : 2026

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The abstract should provide a concise yet comprehensive overview of the study, typically ranging from 150 to 300 words. It begins with a brief introduction that establishes the topic and explains its importance, giving readers immediate context for why the research matters. This is followed by a clear statement of the problem that identifies the central issue and highlights the research gap, pointing out what is lacking, unclear, problematic, or underexplored in existing literature or practice, and explaining how the study intends to address this gap. The abstract should then state the objectives of the study in one or two sentences, ensuring they are specific, measurable, and directly connected to the problem identified. After this, the methods are summarized by briefly describing the research design, participants, instruments, and procedures, with enough detail to show how the study was conducted but without unnecessary elaboration. The results are then presented in a straightforward sentence or two, emphasizing the most significant findings that directly

relate to the objectives and provide evidence to answer the research questions. Finally, the abstract concludes with a sentence that ties the results back to the problem and research gap, showing how the study contributes new knowledge, insights, or practical applications, and indicating its broader implications. After the abstract, a list of 3 to 6 keywords should be added, arranged alphabetically or by relevance, to capture the main themes of the study and aid in indexing for future reference. If this template is not showing/downloaded properly, you may access the template at https://tinyurl.com/COEAManuscriptTemplate2.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Bridges are critical components of transportation infrastructure, serving as lifelines for economic activity, emergency response, and regional connectivity. Ensuring their safety and long-term performance is essential, particularly in the Philippines, which is highly exposed to natural hazards such as earthquakes, typhoons, flooding, and soil erosion. These environmental and operational stresses accelerate structural deterioration, making continuous monitoring increasingly important (Brownjohn, 2007; Farrar & Worden, 2007).

This need is not merely theoretical for Region 2. Within the span of a single year, Cagayan Valley experienced two major bridge failures that exposed both the structural risks of relying on periodic inspection and the steep economic cost of losing a river crossing. On February 27, 2025, a span of the newly opened Cabagan–Sta. Maria Bridge in Isabela — a ₱1.2-billion, 990-meter structure connecting Cabagan and Sta. Maria — collapsed after an overloaded dump truck crossed it, injuring six people and severing a key route across the Cagayan River for nearly a year (Philstar.com, 2025a; Philippine Daily Inquirer, 2026). DPWH's initial investigation pointed to the bridge's design load parameters, which allowed only light vehicles, as a contributing factor in the collapse (Rappler, 2025a). Following the collapse, the Senate's blue ribbon committee opened an inquiry, citing significant economic losses alongside the casualties as justification for a broader review of DPWH contracting practices (Philippine Daily Inquirer, 2025c).

Then, on October 6, 2025, the 45-year-old Piggatan Bridge in Alcala, Cagayan, gave way as multiple overloaded trucks carrying palay and corn crossed it during harvest season, injuring seven people and cutting off the route linking Alcala to Tuguegarao City and other Cagayan trading hubs (Philstar.com, 2025b; Rappler, 2025b). Local officials estimated that around 100,000 farmers in Alcala and neighboring towns depended on that crossing to move their produce, and with the detour stretching a ten-minute trip into an hour-long one, growers reported spoiled harvests and uncertain income even as they faced rising fuel and transport costs (Philippine Daily Inquirer, 2025b).

These recent collapses have also raised concern over older structures still in daily use, chief among them the Buntun Bridge in Tuguegarao City. Opened in 1969 and stretching 1,369 meters across the Cagayan River, Buntun Bridge was for decades recognized as the longest river bridge in the Philippines — a title it held until the opening of the newer Camalaniugan Bridge in January 2026 — and it remains the sole direct connection between Tuguegarao City, Cagayan's provincial capital, and Solana, as well as the broader route toward Isabela, Kalinga, and Apayao (Philippine Information Agency, 2025). After 56 years of continuous service, the bridge has shown clear signs of aging: engineers have documented fatigue in its steel truss members, and holes and cracks have repeatedly appeared in its concrete deck, prompting emergency repairs as early as 2023 (Philippine Information Agency, 2025). In March 2025, weeks after the Cabagan–Sta. Maria collapse, DPWH imposed a 15-ton load restriction on Buntun Bridge in response to worsening structural conditions, barring trucks, trailers, and fully loaded buses (Philippine News Agency, 2025). The bridge has also been temporarily closed outright during major

flood events, including during Typhoon Uwan in November 2025 and Typhoon Tino shortly after, when rising Cagayan River water levels forced authorities to shut it down for hours at a time (Philippine Daily Inquirer, 2025a; Manila Bulletin, 2025). Because Buntun Bridge is the only crossing linking Tuguegarao City to Solana and the towns beyond, each closure or restriction disrupts commuting, emergency response, and the movement of goods for the entire provincial capital, with logistics and delivery operations across the region affected until the bridge reopens. Given its critical role, its documented deterioration, and the load and weather-based restrictions already being managed reactively rather than through continuous data, Buntun Bridge presents both an urgent monitoring need and a practical, accessible site for testing a real-time frequency-based SHM system under real operating conditions.

Together, these incidents illustrate how deterioration and overloading can go undetected between visual inspections, and how the consequences — injuries, severed regional connectivity, and cascading economic losses to farmers, businesses, and commuters — fall directly on communities in Region 2. They underscore the urgency of a monitoring approach that can track a structure's condition continuously rather than periodically, potentially flagging structural stress before it results in both human and economic loss.

Structural Health Monitoring (SHM) has emerged as a modern engineering approach for assessing the condition of structures through continuous or periodic measurement of structural responses (Brownjohn, 2007; Farrar & Worden, 2007). Among various SHM techniques, frequency-based monitoring is widely recognized as a reliable

method for detecting structural changes (Avci et al., 2021; Azhar et al., 2024). The natural and resonant frequencies of a structure are directly related to its mass and stiffness; therefore, any damage such as cracks, corrosion, or joint degradation can result in measurable shifts in frequency characteristics (Salawu, 1997; Doebling et al., 1996; Carden & Fanning, 2004). In practice, vibration signals are acquired from the structure and processed to extract frequency information using techniques such as Fast Fourier Transform (FFT) and spectral analysis (Cooley & Tukey, 1965; Welch, 1967). These methods allow identification of dominant frequency components even in noisy environments, making them suitable for real-world applications (Hasani & Freddi, 2023; Rainieri & Fabbrocino, 2014).

Despite the effectiveness of frequency-based SHM, existing systems in the Philippines are largely dependent on imported technologies, including high-cost accelerometers, data acquisition systems, and proprietary software (Komarizadehasl et al., 2022a; Hidalgo-Fort et al., 2024). These systems are often expensive and inaccessible for widespread deployment, limiting their use to large-scale infrastructure projects (Giermann et al., 2025). As a result, many bridges — including those in Region 2 — rely primarily on periodic visual inspections, which, as the Cabagan–Sta. Maria, Piggatan, and Buntun Bridge cases demonstrate, may fail to detect early-stage structural deterioration or flag structures approaching their load limits before catastrophic failure occurs.

To address this gap, this study proposes the development of a localized, real-time frequency monitoring system for Structural Health Monitoring using MEMS-based Inertial Measurement Unit (IMU) sensors and an ESP32-S3 microcontroller (Komarizadehasl et al., 2022b; Hidalgo-Fort et al., 2024). The system acquires vibration signals through MEMS

IMUs, processes the data using FFT-based techniques, and extracts natural and resonant frequencies in real time (Cooley & Tukey, 1965; Welch, 1967). A web-based interface is also implemented for visualization and monitoring, following a similar approach to prior Philippine efforts in low-cost bridge sensor networks (Gaviña et al., 2017; Payawal et al., 2017; Lau et al., 2018). The system is considered localized as it utilizes cost-efficient, locally accessible components and is designed to be adaptable to Philippine bridge conditions, and will be tested on Buntun Bridge in Tuguegarao City as a representative, high-criticality structure in Region 2. By focusing on affordability, portability, and real-time capability, the study aims to contribute to the development of practical SHM solutions for resource-limited environments (Komarizadehasl et al., 2022a; Philippine Council for Industry, Energy and Emerging Technology Research and Development, n.d.) — and to a monitoring approach that could have flagged the structural stress behind incidents like the Cabagan–Sta. Maria and Piggatan bridge collapses before they occurred.

1.2 Statement of the Problem

Bridge infrastructure in Cagayan Valley is managed primarily through periodic visual inspection and reactive intervention — as seen in the Cabagan–Sta. Maria and Piggatan Bridge failures, where structural stress went undetected until collapse. Buntun Bridge, Tuguegarao City's sole direct crossing to Solana, is currently under a load restriction imposed without any instrumented record of its vibrational behavior. Existing frequency-based SHM technologies remain largely imported and cost-prohibitive, limiting deployment to large-scale projects and leaving structures like Buntun Bridge without a documented reference for detecting future structural change. This study addresses that gap by developing

a localized, low-cost, real-time frequency monitoring system and using it to establish an internal baseline of Buntun Bridge's dominant vibration frequencies under normal traffic conditions. Specifically, this study seeks to answer the following questions:

1. How can a real-time frequency monitoring system be developed to acquire and process structural vibration data?
2. How effectively can dominant structural frequencies be extracted from vibration signals under varying excitation and noise conditions?
3. How can the system provide real-time monitoring and visualization of structural vibration characteristics?

1.3 Objective of the Study

This study aims to develop and deploy a localized, real-time frequency monitoring system for Structural Health Monitoring, and to use it to establish a baseline of dominant structural frequencies for Buntun Bridge, using MEMS-based IMU sensors and an ESP32-S3 microcontroller. Specifically, it aims to:

1. Develop a real-time monitoring system capable of acquiring and processing structural vibration data from Buntun Bridge.
2. Extract and establish a baseline range of dominant structural frequencies from acquired vibration signals under varying traffic conditions.
3. Develop a real-time visualization interface for displaying structural vibration characteristics and frequency behavior.

1.4 Conceptual or Theoretical Framework of the Study

This study is grounded in the principle that a structure's dynamic behavior — its natural and resonant frequencies — is directly governed by its mass and stiffness. Any change in a structure's physical condition, such as cracking, corrosion, or joint degradation, alters its stiffness and therefore produces a measurable shift in these frequency characteristics. This relationship is the theoretical basis for frequency-based Structural Health Monitoring: by continuously observing a structure's frequency behavior relative to its own established baseline, meaningful changes in structural condition can, in principle, be identified without requiring physical inspection.

The conceptual framework of the developed system follows an Input–Process–Output (IPO) model, illustrated in Figure 1.

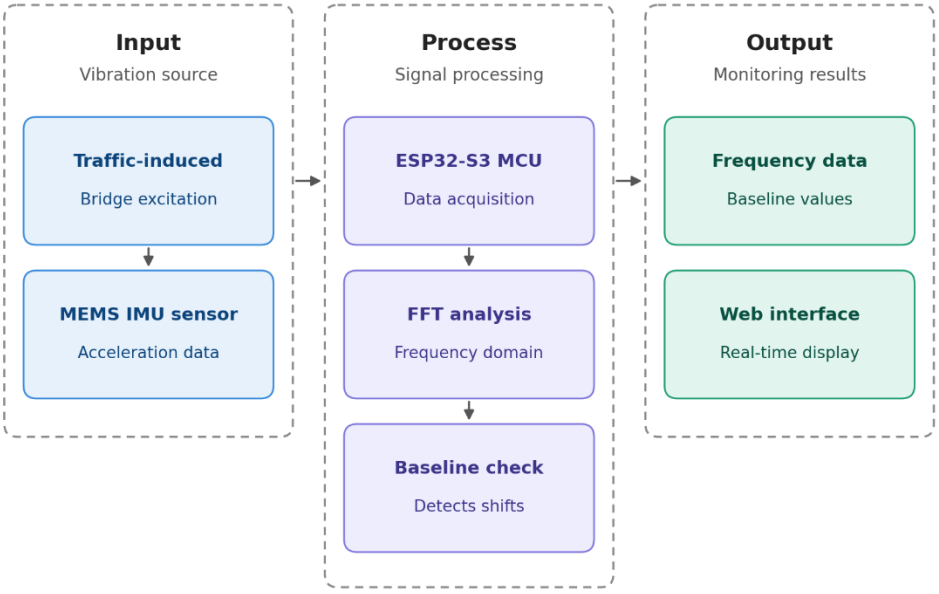


Figure 1. *Input–Process–Output (IPO) Conceptual Framework of the SHM System*

In the Input stage, vibration is generated by traffic-induced excitation as vehicles cross the bridge, and this vibration is captured as raw acceleration data by MEMS-based IMU sensors mounted on the structure.

In the Process stage, the ESP32-S3 microcontroller acquires and preprocesses the incoming sensor data, applies the Fast Fourier Transform to convert the signal from the time domain to the frequency domain, and compares the resulting dominant frequencies against the baseline frequency range established for the tested structure.

In the Output stage, the extracted frequency data and its relation to the baseline are transmitted to a web-based interface, providing real-time visualization of the structure's vibration characteristics for monitoring purposes.

This framework positions the study's contribution specifically within the Process and Output stages — developing the acquisition, frequency-extraction, and visualization pipeline — while the Input stage, the bridge structure's actual vibration response, is the real-world phenomenon the system is designed to observe.

1.5 Significance of the Study

This study contributes to the development of a cost-effective and accessible Structural Health Monitoring solution for bridge infrastructure in the Philippines. By utilizing MEMS-based IMU sensors and an ESP32-S3 microcontroller, the system offers a practical, locally deployable alternative to the expensive imported SHM technologies that have limited continuous monitoring to only a small number of large-scale infrastructure projects.

To residents and commuters in Tuguegarao City and Solana, particularly those who rely on Buntun Bridge as their only direct crossing, this study represents a step toward monitoring the condition of a structure the community depends on daily for travel, trade, and emergency access — reducing reliance on reactive measures such as load restrictions and emergency closures imposed only after visible damage or a critical event occurs.

For engineers and infrastructure inspectors, the system offers a supplementary tool for continuous, real-time observation of structural vibration behavior, providing data between scheduled visual inspections rather than relying on inspection intervals alone.

For DPWH and other government agencies responsible for bridge maintenance, the system supports more informed, data-driven decisions on monitoring priorities and maintenance timing, offering a low-cost option that could extend structural oversight to bridges — especially aging or heavily trafficked ones like Buntun Bridge — that would otherwise go unmonitored between inspections.

For the local economy and communities in Region 2, including farmers, traders, and businesses whose livelihoods depend on uninterrupted access across the Cagayan River, this study supports the broader goal of reducing the risk of unplanned bridge closures and the trade, transport, and income disruptions that follow, as seen in the Cabagan–Sta. Maria and Piggatan Bridge incidents.

For the academic and engineering research community, this study contributes a documented reference for the development of embedded and IoT-based SHM systems using locally accessible components, offering a foundation that future researchers can build on —

whether by extending the system to other structures, incorporating additional sensing modalities, or advancing it toward damage classification and predictive capabilities.

Ultimately, the localized design of this study promotes technological independence in structural monitoring and demonstrates that meaningful SHM capability can be achieved using affordable, readily available components — encouraging further innovation in this space within resource-limited settings.

1.6 Scope and Delimitation

This study focuses on the development and evaluation of a localized, frequency-based Structural Health Monitoring system for bridge structures. The system is designed to acquire vibration signals using MEMS-based IMU sensors, process the signals through FFT, and extract frequency characteristics for real-time monitoring. The hardware system utilizes an ESP32-S3 microcontroller for data acquisition and processing, and the extracted frequency data are transmitted to a web-based interface for real-time visualization.

The system will be tested on the Buntun Bridge in Tuguegarao City, Cagayan, as the representative structure for this study. Data collection is limited to this single bridge site; the system's performance and extracted frequency characteristics are specific to Buntun Bridge and are not claimed to generalize to bridges of different structural types, spans, or material conditions without further testing.

Since no prior instrumented baseline exists for Buntun Bridge, this study establishes its baseline frequency ranges using the developed device itself, based on vibration data gathered under varying vehicular traffic conditions. The resulting baseline is therefore

internal to the system and this deployment, and is not validated against an external reference sensor or a separately verified ground truth.

The study is limited to frequency-based monitoring and does not include structural diagnosis, damage classification, predictive modeling, load-capacity analysis, or repair recommendations. Interpretation of any observed frequency shift as a specific damage type or severity level is outside the scope of this study.

The system is further limited by hardware constraints such as processing speed, memory capacity, and sensor resolution. Environmental factors such as temperature, humidity, and electromagnetic interference are not fully incorporated into the analysis. Data collection is also limited to the specific time periods and traffic conditions observed during testing, and may not capture the full range of seasonal, weather-related, or long-term structural variation.

1.7 Definition of Terms

Baseline Frequency Range: The set of natural and resonant frequency values established for a specific structure under normal operating conditions, used as a reference point against which future vibration readings can be compared.

Excitation: An external force or disturbance, such as vehicular traffic, wind, or ambient vibration, that causes a structure to vibrate and produce a measurable dynamic response.

ESP32-S3*:* A microcontroller used for data acquisition, processing, and wireless communication in the system.

Fast Fourier Transform (FFT)*:* A signal processing technique used to convert time-domain signals into frequency-domain data.

Frequency Monitoring*:* The process of analyzing vibration signals to extract frequency characteristics related to structural behavior.

Localized (System)*:* In this study, refers to a system designed using cost-efficient, locally accessible components rather than imported, proprietary hardware, making it more feasible to deploy and maintain in resource-limited settings.

MEMS IMU (Micro-Electro-Mechanical Systems Inertial Measurement Unit)*:* A sensor that measures acceleration and motion, used in this study for vibration data acquisition.

Natural Frequency*:* The inherent frequency at which a structure vibrates when disturbed, determined by its mass and stiffness.

Real-Time Monitoring*:* Continuous or near-instantaneous observation and processing of data.

Resonant Frequency*:* The frequency at which a structure experiences maximum vibration amplitude due to external excitation.

Structural Health Monitoring (SHM)*:* A method of evaluating the condition of a structure through measurement and analysis of its response.

Vibration Signal*:* The raw time-domain data captured by the sensor, representing the structure's motion or acceleration over time, prior to frequency-domain processing.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter reviews literature and related studies relevant to the development of a localized, real-time frequency monitoring system for bridge Structural Health Monitoring (SHM). The review is organized thematically to connect the study with established work on vibration-based SHM, structural dynamic properties, operational modal analysis, frequency-domain signal processing, low-cost MEMS sensing, real-time and Internet of Things (IoT)-enabled monitoring, and the effects of environmental and operational variability. International and Philippine studies are considered to identify established practices, practical limitations, and the specific gap addressed by the present study.

2.1. Introduction

Structural Health Monitoring has developed from periodic condition assessment toward measurement-based approaches capable of observing structural response during service. For bridge structures, vibration measurements are particularly useful because dynamic behavior can be observed without requiring the application of a controlled test load. Contemporary reviews describe bridge SHM as a combination of sensing, data acquisition, signal processing, modal identification, and decision-support methods that can operate continuously or periodically depending on the monitoring objective (Azhar et al., 2024; Chaudhari et al., 2026). The literature therefore provides a technical basis for a field-deployed system that measures bridge vibration, converts the response into frequency-domain information, and presents the extracted parameters for monitoring.

The reviewed literature also shows that frequency-based monitoring is not a simple matter of obtaining a spectral peak and treating every change as damage. Modal frequencies are influenced not only by structural condition but also by temperature, humidity, wind, traffic loading, boundary conditions, sensor characteristics, and measurement noise (Sohn, 2007; Wang et al., 2022). This distinction is important for the present study because the intended baseline is to be established on an operating bridge under different vehicular traffic conditions. The literature consequently supports the use of repeated observations and explicit characterization of operational variability rather than a single reading taken under one field condition.

2.2. Vibration-Based Structural Health Monitoring of Bridges

2.2.1 Structural Health Monitoring and Dynamic Response

Structural Health Monitoring refers to the systematic observation of a structure through measured responses and the analysis of those responses to identify changes in behavior, condition, or performance. Vibration-based SHM is one of the principal approaches because structural vibration carries information about dynamic properties that depend on the physical characteristics of a structure. Reviews of vibration-based damage detection show that changes in modal properties, including natural frequencies and mode shapes, have long been used as damage-sensitive features because alterations in mass, stiffness, or boundary conditions can modify the dynamic response (Das et al., 2016; Avci et al., 2021).

For bridges, vibration monitoring is particularly suitable for in-service observation because ambient and operational excitation from traffic, wind, and other sources can provide measurable structural response without the need for a dedicated excitation device. Hasani and Freddi (2023) describe Operational Modal Analysis (OMA) as a practical framework for extracting modal information from output measurements under natural or operational loading. This makes OMA relevant to field situations in which a bridge cannot be closed for controlled excitation and in which the objective is to understand the bridge response during normal service.

The literature nevertheless distinguishes monitoring from diagnosis. A frequency change may indicate that the dynamic behavior has changed, but a frequency shift alone generally does not identify a unique damage mechanism or severity. Reference-free damage-detection research emphasizes that reliable diagnosis is more difficult when an intact reference state is unavailable and when environmental or operational effects overlap with damage-related changes (Moravvej & El-Badry, 2024). This distinction supports the present study's limitation to frequency monitoring and baseline characterization rather than structural damage classification.

2.2.2 Natural Frequency, Modal Parameters, and Structural Properties

The dynamic properties of a structure are governed by its mass, stiffness, and boundary conditions. In a simplified single-degree-of-freedom representation, the natural frequency is related to stiffness and mass through the square-root relationship between those parameters. In multi-degree-of-freedom bridges, the same general principle extends to a set of modal frequencies and corresponding mode shapes. Consequently, changes in structural

properties may produce measurable changes in modal parameters. This is the fundamental reason that vibration-based SHM can use modal information as an indirect indicator of structural change (Wang et al., 2022; Azhar et al., 2024).

It is important, however, to distinguish natural frequency from resonant response. Natural frequency is an inherent dynamic property associated with a mode of the structure, whereas resonance describes a condition in which excitation strongly amplifies the response near a modal frequency. In field measurements, the prominent spectral peaks observed under operational excitation are therefore interpreted as candidate modal or dominant frequency components rather than as an automatic measurement of every theoretical natural frequency. This distinction becomes especially important when the excitation spectrum is uneven, when modes are closely spaced, or when the signal-to-noise ratio is low (Chen et al., 2017; Hasani & Freddi, 2023).

2.3. International Perspective

2.3.1 Operational Modal Analysis and Frequency-Domain Identification

Operational Modal Analysis estimates modal characteristics from structural response measurements while the structure is subjected to ambient or operating loads. For bridges, common output-only identification methods include Peak Picking (PP), Frequency Domain Decomposition (FDD), and Stochastic Subspace Identification (SSI). Hasani and Freddi (2023) report that PP is the simplest frequency-domain approach and identifies candidate modal frequencies from peaks in response spectra. FDD extends frequency-domain analysis by decomposing the spectral density matrix, allowing more complex modal

behavior to be separated. SSI operates in the time domain and can provide robust modal identification and mode shapes, although it is more computationally demanding.

Chen et al. (2017) demonstrated the practical value and limitations of these approaches on an eleven-span concrete bridge under weak ambient excitation. Using inexpensive accelerometers, they identified multiple modal frequencies below 10 Hz despite weak excitation and measurement noise. Their comparison showed that Peak Picking could identify a large number of frequencies efficiently, while SSI produced higher-quality mode-shape information at higher computational cost. This finding is directly relevant to an embedded real-time system: a lightweight peak-picking approach can be reasonable for preliminary or continuous monitoring when computational resources are limited, provided that the limitations of the method are explicitly acknowledged.

The literature further indicates that frequency-domain peak identification is most dependable when modal peaks are reasonably separated and the measured response has adequate signal-to-noise ratio. Closely spaced modes, harmonics, local vibration sources, and noise can create ambiguous peaks or shift the apparent maximum. For that reason, the present study should treat the selected spectral peak as a dominant frequency estimate generated by a defined processing procedure, rather than claiming that simple peak picking is universally equivalent to full modal identification (Hasani & Freddi, 2023; Azhar et al., 2024).

2.3.2 Fast Fourier Transform and Vibration Signal Processing

The Fast Fourier Transform (FFT) is an efficient algorithm for computing the Discrete Fourier Transform and is widely used to transform sampled vibration signals from

the time domain into a frequency-domain representation. In SHM, this transformation allows the energy distribution of a measured acceleration signal to be examined across frequency. Once the spectrum is formed, candidate dominant or modal frequencies can be selected from spectral peaks, subject to the resolution and quality of the underlying measurement (Hasani & Freddi, 2023).

Field vibration analysis generally requires preprocessing before FFT-based interpretation. Detrending and removal of offset reduce contamination associated with nonstationary mean components, while filtering can be used when the measurement contains strong out-of-band noise. Sampling frequency also constrains the highest frequency that can be represented without aliasing, and the record length affects the spacing of discrete frequency bins. These considerations make the signal-processing chain an integral part of the monitoring method rather than a secondary implementation detail. Chen et al. (2017) and Hasani and Freddi (2023) both show that preprocessing and modal-identification choices affect the reliability of frequency estimates obtained from in-situ bridge responses.

For an embedded system, FFT processing must also be balanced against processor capacity, memory, latency, and the desired update rate. Low-cost edge-computing SHM systems demonstrate that spectral or modal processing can be performed close to the sensor node, reducing the amount of raw data that must be transmitted and enabling near-real-time monitoring (Hidalgo-Fort et al., 2024). This supports the use of an ESP32-S3-based architecture in which vibration acquisition and frequency extraction occur locally before selected results are sent to a web interface.

2.3.3 Low-Cost MEMS Accelerometers for Bridge Monitoring

The increasing availability of MEMS accelerometers has reduced the cost and physical size of vibration-sensing hardware. Research on low-cost accelerometer systems indicates that such sensors can be used for bridge vibration measurement and eigenfrequency estimation when their noise, sampling behavior, synchronization, and installation conditions are adequately considered. Komarizadehasl et al. (2022a) developed a low-cost triaxial accelerometer system and demonstrated its use for eigenfrequency analysis of a footbridge, while later work by the same research group showed that low-cost wireless accelerometers could support automated field data collection and operational modal analysis (Komarizadehasl et al., 2022b).

More recent field evidence continues to support the feasibility of inexpensive MEMS sensing while emphasizing its limitations. Giermann et al. (2025) tested low-cost MEMS acceleration sensors both under laboratory conditions and on a bridge, where a microcontroller and wireless communication were used to make measurements available online. The study also observed random and systematic influences that caused measurement deviations. These findings reinforce the need for repeatability checks, stable installation, and transparent reporting of sensor limitations when a low-cost platform is used in place of a calibrated commercial monitoring system.

The recent literature also shows that MEMS sensing is moving beyond laboratory prototypes toward practical bridge deployments. Hasani et al. (2024) described a wireless data acquisition system for bridge OMA using MEMS accelerometers and a web-based interface, while Giermann et al. (2025) reported long-term bridge deployment of a low-cost

MEMS system. Together, these studies support the feasibility of a compact sensor-microcontroller architecture for field monitoring, but they also highlight that inexpensive sensing should be evaluated in terms of the specific frequency band, noise floor, sampling rate, synchronization, and installation configuration required by the target bridge.

2.3.4 Real-Time, Wireless, and IoT-Enabled SHM

Real-time SHM increasingly combines sensing, embedded computation, wireless communication, and web-based visualization. Such architectures allow sensor nodes to preprocess measurements locally and transmit summaries or selected features to a remote interface. Hidalgo-Fort et al. (2024) presented a low-cost, low-power edge-computing SHM architecture for bridges in which processing is performed at the monitoring node and the wireless network is used for communication. The system was deployed on the Eduardo Torroja Bridge and produced modal-frequency results with an error below 1.72% when compared with calibrated commercial systems.

Hasani et al. (2024) similarly integrated MEMS sensing, wireless communication, synchronization, data preprocessing, and a web-based graphical interface into a bridge monitoring architecture. These studies establish that the monitoring problem is not limited to sensor measurement alone; reliable SHM requires a complete data path from acquisition through processing, transmission, storage, and visualization. The present study adopts this system-level perspective, but limits its core analysis to vibration-frequency characterization rather than full-scale automated diagnosis.

A 2026 systematic review by Chaudhari et al. (2026) further identifies low-cost MEMS sensing, IoT connectivity, and data-driven modal analysis as important directions

in bridge SHM. At the same time, the review emphasizes that operational and environmental variability remains a major challenge because it can conceal or mimic structural changes. This reinforces the need for the present study to establish its frequency baseline under explicitly documented operating conditions rather than relying on an undifferentiated pool of field readings.

2.3.5 Environmental and Operational Variability in Frequency Monitoring

One of the most important findings across bridge SHM literature is that modal frequency is not perfectly constant during service. Environmental and operational effects such as temperature, humidity, wind, vehicle loading, traffic intensity, and changing boundary conditions can alter the measured modal characteristics. Sohn (2007) established environmental and operational variability as a central issue in SHM, and Wang et al. (2022) later reviewed methods for predicting and removing these effects from modal-frequency measurements.

Traffic is especially relevant to the present study because it is both the primary operational excitation of the target bridge and a source of variability in the measured response. Zhou and Sun (2019), using long-term monitoring of the Donghai Bridge, demonstrated that traffic and temperature contribute to modal-frequency variation and that such effects can mask changes associated with deterioration. Earlier studies likewise documented changes in measured bridge dynamic characteristics associated with vehicle mass and parked vehicles (Kim et al., 2003; Chang et al., 2014). Thus, differences in traffic condition should not simply be treated as unwanted noise; they are an important operational variable that must be characterized when establishing a field baseline.

For the present study, this literature suggests a practical interpretation of traffic conditions. Rather than claiming that a single arbitrary vehicle count defines a universal threshold for "varying traffic conditions," the study can operationally define sessions by observable traffic states and document the conditions during acquisition. A minimum comparison between peak and off-peak periods provides two distinguishable operational states, while weekday versus weekend measurements can provide an additional contrast when feasible. The resulting baseline should therefore be described as a condition-specific empirical baseline for the monitored bridge and monitoring period, not as an immutable structural constant.

2.4. National/Local Perspective

2.4.1 Philippine Applications of Sensor-Based Bridge SHM

Philippine research demonstrates that sensor-based SHM and remote bridge monitoring are feasible in local infrastructure settings. Gaviña et al. (2017) developed and applied SmartBridge sensor nodes to existing concrete bridges with AASHTO Type IV girders. The system used acceleration measurements, wireless communication, and an online platform for monitoring, with emphasis on appropriate sensor installation and placement. This study is significant to the present research because it shows an earlier Philippine effort to move bridge assessment beyond periodic visual inspection toward instrumented monitoring and remote data access.

Payawal et al. (2017) investigated MEMS-based accelerometer measurements through remote monitoring of the Padre Jacinto Zamora Flyover and compared actual sensor

readings with theoretical structural-model outputs. Their work illustrates a practical issue also relevant to the present study: field measurements from low-cost sensors may differ substantially from idealized analytical predictions, making calibration and field characterization necessary rather than assuming that a theoretical value is automatically equivalent to an observed one.

Lau et al. (2018) extended sensor-based bridge monitoring in the Philippine context through a fuzzy-based structural health rating procedure for Zamora Bridge. Three-dimensional acceleration measurements were obtained from sensors installed on structural components and used in an assessment framework. Although the analytical approach differs from the present study, the work demonstrates that local bridge SHM research has already explored acceleration sensing as a supplement to visual inspection and that sensor-derived measurements can be incorporated into condition-assessment workflows.

2.4.2 Philippine Real-Time Monitoring and Institutional Developments

The Philippine Council for Industry, Energy and Emerging Technology Research and Development (PCIEERD) has also supported the Universal Structural Health Evaluation and Recording System (USHER), a system combining accelerographic sensing, a web portal, and a mobile application. The project describes 24/7 monitoring and natural-frequency recording intended to support structural assessment before, during, and after strong earthquakes. It also identifies the limitations of visual inspection as a primary monitoring practice. The existence of this project indicates that real-time, web-connected structural monitoring is already being pursued in the Philippines, although its principal

focus and sensing architecture differ from the present bridge-frequency baseline study (PCIEERD, n.d.).

Recent conditions surrounding Buntun Bridge further strengthen the practical relevance of continuous or repeated data acquisition. Official Philippine reporting documented deterioration-related load restrictions on the bridge in 2025, including a 15-ton limit for heavy trucks, while the Philippine Information Agency described fatigue in some steel members and cracks in the concrete deck during the bridge's continuing rehabilitation (Philippine News Agency, 2025; Philippine Information Agency, 2025). These reports do not establish a vibration-based diagnosis and should not be interpreted as evidence of a frequency anomaly. Their relevance to this study is instead contextual: the bridge is an aging, heavily used transportation structure for which additional quantitative observations of dynamic behavior may be useful as a supplementary monitoring layer.

2.4.3 Local Research Gap and Relevance to Buntun Bridge

The Philippine studies reviewed above show a progression from acceleration sensing and wireless data transmission toward remote and real-time monitoring. However, the specific combination proposed in the present study remains distinct in emphasis: an embedded, low-cost MEMS-based system is used to obtain field vibration data on Buntun Bridge, transform the measurements into the frequency domain, extract dominant frequency values for repeated sessions, and establish an empirical baseline while explicitly considering traffic conditions. The reviewed local works primarily emphasize acceleration measurement, sensor placement, theoretical calibration, fuzzy-based assessment, or

earthquake-oriented monitoring rather than a traffic-conditioned dominant-frequency baseline for an aging steel truss bridge.

This gap should be stated carefully. The literature review does not establish that no other Philippine project has ever performed traffic-conditioned frequency monitoring. Rather, the sources identified for this study do not provide a directly equivalent field methodology combining low-cost MEMS sensing, embedded FFT-based dominant-frequency extraction, repeated traffic-condition sessions, and an empirical baseline for Buntun Bridge. The present study therefore contributes an applied field implementation that connects established SHM concepts with a resource-conscious monitoring architecture suited to a specific Philippine bridge.

2.5. Synthesis

The reviewed literature establishes four closely connected findings that guide the present study. First, vibration-based SHM is a recognized approach for observing changes in bridge dynamic behavior, with modal frequency among the commonly used features for structural characterization (Das et al., 2016; Azhar et al., 2024). Second, operational modal analysis allows modal information to be estimated from responses under traffic and other ambient loads, making field deployment practical when controlled excitation is not available (Chen et al., 2017; Hasani & Freddi, 2023). Third, low-cost MEMS accelerometers and embedded wireless systems have demonstrated practical potential for bridge monitoring, although noise, synchronization, installation, and calibration remain important limitations (Komarizadehasl et al., 2022a; Komarizadehasl et al., 2022b; Hasani et al., 2024; Giermann et al., 2025). Fourth, traffic and environmental conditions can change

measured modal frequencies sufficiently to obscure structural changes, making condition-aware baseline construction essential (Sohn, 2007; Zhou & Sun, 2019; Wang et al., 2022).

These findings directly shape the methodological position of the present study. FFT-based frequency extraction and peak picking are appropriate as a computationally lightweight means of obtaining a repeatable dominant-frequency indicator, but the indicator should be interpreted within the limits of the method and the measurement conditions. Because the target bridge is monitored in service, the baseline should be formed from repeated field sessions rather than from a single test or an assumed theoretical frequency. Traffic conditions should be explicitly documented and compared, with peak versus off-peak operation providing the minimum practical distinction and weekday versus weekend observations adding an additional contrast when feasible. The empirical baseline is therefore not intended to define an absolute structural safety threshold; rather, it provides a reference range of observed frequency behavior for the monitored bridge and study period.

The literature also supports a clear boundary for interpretation. A statistically significant difference among traffic conditions would demonstrate that operating condition is associated with changes in the measured dominant frequency, but it would not by itself prove structural damage. Likewise, a frequency value outside the baseline would justify attention or further investigation, but it would not identify a specific defect without additional evidence. This conservative interpretation is consistent with the broader SHM literature, which treats environmental and operational effects as sources that must be separated from damage-related changes before making structural-health conclusions (Wang et al., 2022; Chaudhari et al., 2026).

Overall, the reviewed studies justify the development of a localized real-time frequency monitoring system using MEMS sensors, embedded processing, FFT-based spectral analysis, and web visualization. They also reveal the central research need addressed by this thesis: translating the general feasibility of low-cost vibration monitoring into a documented, field-tested procedure for establishing and interpreting a traffic-aware dominant-frequency baseline on an operating Philippine bridge. This synthesis provides the basis for the research design, data-gathering protocol, repeatability checks, baseline construction, and statistical comparison to be presented in Chapter III.

CHAPTER III

METHODOLOGY

This chapter presents the methodology for the development and field evaluation of the localized, real-time frequency monitoring system for bridge Structural Health Monitoring (SHM). The study combines embedded vibration acquisition, frequency-domain signal processing, repeated field measurements, empirical baseline construction, and statistical comparison of dominant frequency under documented traffic conditions. The procedure is designed to be reproducible and to remain within the stated scope of the study: characterization of vibration-frequency behavior rather than structural damage diagnosis or load-capacity assessment.

3.1 General Overview

The study employs a developmental and quantitative field-testing approach. The developmental component covers the design, integration, and implementation of the vibration-monitoring hardware, embedded processing, wireless communication, and web-based visualization interface. The quantitative field-testing component evaluates whether the completed system can repeatedly acquire vibration data on an operating bridge and extract a consistent dominant-frequency indicator under different vehicular traffic conditions.

The overall methodological sequence consists of: (1) securing site permission and safety clearance; (2) preparing and checking the monitoring hardware and software; (3) installing the MEMS-based inertial measurement unit (IMU) on the selected bridge

location; (4) conducting repeated vibration-acquisition sessions under predefined traffic conditions; (5) preprocessing and transforming the vibration signals using the Fast Fourier Transform (FFT); (6) selecting the dominant spectral frequency using a consistent peak-picking rule; (7) aggregating valid session results to establish the empirical baseline; (8) evaluating repeatability using descriptive statistics and coefficient of variation; and (9) comparing frequency distributions across traffic conditions using one-way analysis of variance (ANOVA), with a nonparametric alternative when its assumptions are not adequately supported.

3.2 Study Area

The study will be conducted on Buntun Bridge in Tuguegarao City, Cagayan. The bridge is selected as the field test site because it is an operating transportation structure with continuous vehicular activity and because the present study specifically seeks to evaluate the monitoring system under actual service excitation rather than under laboratory-only conditions. Data collection will therefore be conducted without intentionally introducing abnormal structural loading or artificial excitation.

The exact sensor mounting location, bridge member used as the mounting point, direction of the sensor axes, and the accessible working area will be finalized in coordination with the responsible bridge authority. A site sketch and/or photographic documentation will be included in the final manuscript to identify the sensor location and field setup. Written authorization for access, measurement, and safety procedures shall be

secured before formal data collection begins. Verbal coordination received before written approval shall not be treated as the final authorization for field deployment.

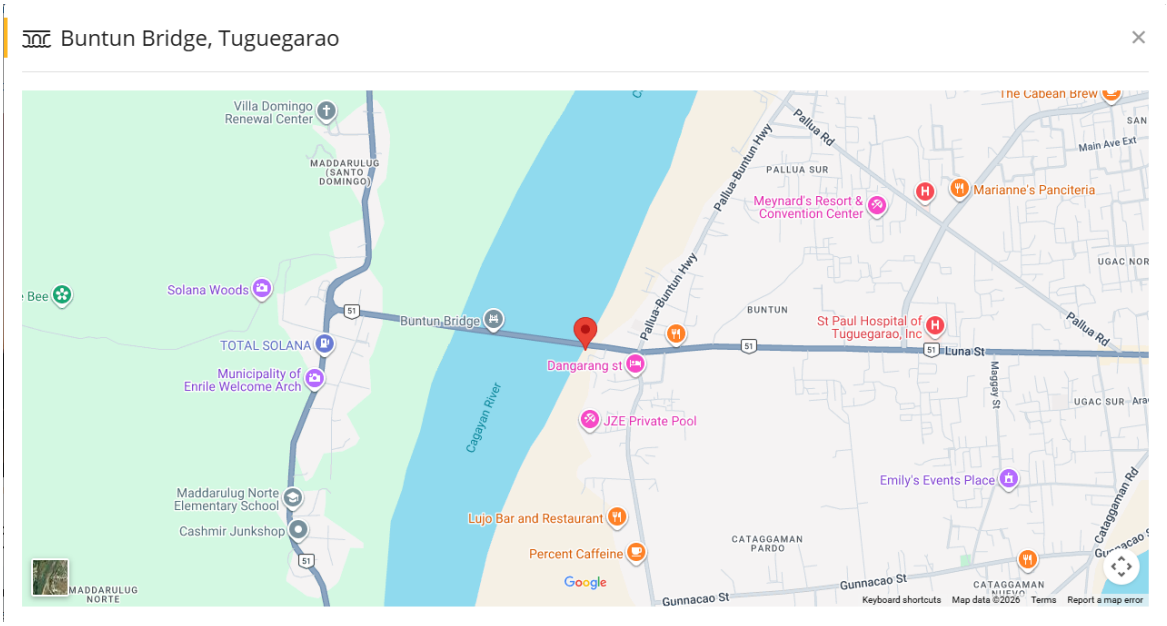


Figure 1. *Map of Buntun Bridge, Tuguegarao*

3.3 Research Design

The study uses a single-site, repeated-measures field design. Each data-collection session is treated as a measurement occasion in which the monitoring system records bridge vibration for a fixed duration while the prevailing traffic condition is documented. The principal response variable is the dominant spectral frequency extracted from the vibration signal. The principal explanatory variable is traffic condition, operationalized into predefined field categories.

The field design is observational with respect to traffic excitation because the researchers do not control the number, type, speed, or weight of vehicles crossing the bridge.

Instead, the study records the bridge response produced by naturally occurring operational loading. This approach is consistent with output-only or operational modal testing, in which vibration responses are collected while a structure remains in service (Hasani & Freddi, 2023).

Because the study does not introduce controlled damage or a known reference excitation, the results are interpreted as an empirical description of frequency behavior for the monitored bridge and study period. A measured frequency outside the established baseline is treated as an observation requiring attention or further investigation, not as proof of a particular damage mechanism.

3.4 System Architecture and Materials

The monitoring system consists of four functional stages: sensing, embedded acquisition and processing, wireless data transmission, and web-based visualization. The MEMS IMU captures structural acceleration in three orthogonal axes. The ESP32-S3 microcontroller samples the sensor output and performs the required preprocessing and FFT operations. The resulting dominant-frequency values and supporting session information are transmitted to the monitoring interface for visualization and logging.

The final manuscript shall report the exact sensor model, number of sensor nodes, mounting hardware, sampling configuration, power source, communication method, data-storage method, and software environment actually used during the completed field trials. Table 1 provides the required reporting structure so that the final configuration is documented rather than inferred.

Component	Function	Key parameter to report	Final configuration
MEMS-based IMU	Measures bridge acceleration/vibration	Sensor model, range, resolution, axes	To be filled from final hardware
ESP32-S3	Data acquisition, preprocessing, FFT, communication	Firmware, sampling rate, processing window	To be filled from final firmware
Mounting assembly	Provides stable mechanical coupling	Mounting point and attachment method	To be documented on site
Wireless/network interface	Transfers processed results	Protocol and update interval	To be filled from final system
Web interface	Displays frequency and session information	Displayed variables and refresh interval	To be documented from final UI
Reference logging	Records session conditions	Date/time, weather, traffic count/category	Field log sheet

Table 1. *Required documentation of the final monitoring-system configuration*

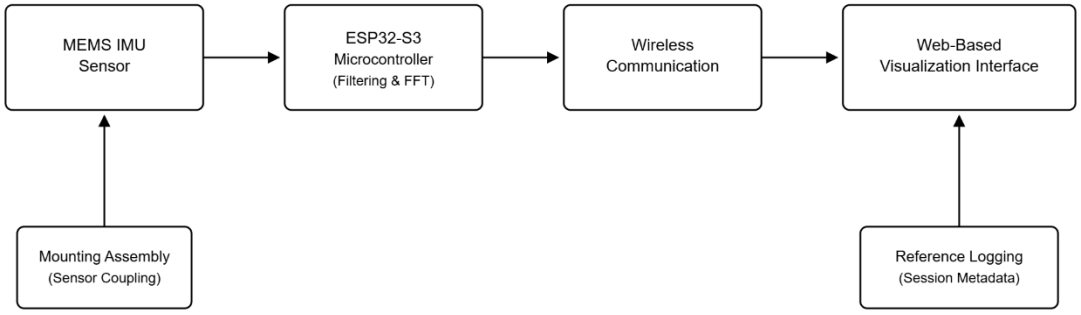


Fig 2. *System Block Diagram*

3.5 Field Measurement Units and Sampling Plan

The unit of observation is a valid measurement session. A session consists of a continuous vibration record of predetermined duration acquired with the same sensor placement, orientation, sampling configuration, and signal-processing settings. The final duration shall be selected during pilot testing to provide sufficient frequency resolution while remaining practical for field operation.

To provide a defensible baseline, the study targets at least five valid sessions for each primary traffic condition, with additional sessions preferred when field access and safety allow. The target is a methodological sampling goal rather than a universal statistical requirement. A session may be rejected before analysis only for predefined technical reasons, such as sensor detachment, severe clipping or saturation, corrupted data, missing timestamps, communication failure that prevents recovery of the required record, or an otherwise documented acquisition error.

The primary comparison will consist of peak-traffic and off-peak sessions. Where field access permits, sessions will also be balanced across weekday and weekend observations so that the baseline is not constructed from only one calendar pattern. Session metadata shall include date, start and end time, traffic category, observed vehicle count or traffic observations, weather notes, and any unusual event that may affect the vibration record.

3.6 Research Instrument and Sensor Installation

The monitoring device serves as the primary research instrument. Before field deployment, the sensor and microcontroller will undergo a functionality check, including power stability, sensor communication, axis response, timestamp continuity, data recording, FFT execution, and web-interface updating. The same configuration will be retained across field sessions unless a documented change is intentionally introduced and reported.

The IMU shall be mounted on a structurally appropriate and accessible bridge member using a mechanically stable attachment method. The mounting surface must provide adequate coupling between the sensor and the bridge member while minimizing movement of the sensor relative to the structure. The sensor axes shall be documented so that the horizontal and vertical directions are known during analysis. The mounting location, orientation, and attachment method shall remain unchanged across comparison sessions whenever practicable.

A zero-offset or static-bias check shall be conducted before each field run, and the device shall be inspected for looseness or cable movement. Where possible, a short stationary record shall be collected before acquisition to establish whether a large offset,

clipping, or abnormal noise is present. These checks support repeatability but do not constitute calibration against a commercial reference instrument.

3.7 Data Gathering Procedure

Formal data collection shall begin only after written permission and site-safety arrangements have been obtained. The researchers shall coordinate with the responsible authority regarding access, allowable working locations, traffic restrictions, and emergency procedures. No activity shall obstruct traffic lanes or require researchers to enter an unsafe traffic path.

1. Pre-field preparation. Charge the monitoring system, verify firmware, check the IMU connection, confirm the sampling and FFT settings, synchronize the system clock, and prepare the field log sheet.
2. Site inspection. Inspect the selected mounting location, identify a safe working position, document environmental conditions, and confirm that the intended sensor attachment can be completed without interfering with bridge operations.
3. Sensor installation. Attach the IMU securely, record its location and orientation, and verify that the accelerometer axes correspond to the intended analysis directions.
4. Stationary check. Record a short stationary segment and verify that the signal is free from obvious saturation, discontinuity, or abnormal noise before starting the session.

5. Traffic-condition observation. Record the scheduled traffic category, actual traffic observations, vehicle count or count rate when feasible, and notable events such as unusually heavy vehicles, stops, congestion, or nearby activities.
6. Vibration acquisition. Start a continuous record for the predetermined session duration. Avoid changing the sensor position or software settings during the session.
7. Session closure. Save the processed frequency result together with the raw or retained time-domain record, session identifier, timestamp, and field-log information. Inspect the record for technical validity before accepting it into the analysis dataset.
8. Repetition. Repeat the procedure across the planned peak and off-peak sessions and, when feasible, across weekday and weekend periods.

3.8 Traffic-Condition Characterization

Traffic condition shall be operationalized using observable field information rather than an arbitrary universal vehicle-count threshold. Two primary operating categories will be used: peak traffic and off-peak traffic. The time windows used for these categories shall be fixed before formal analysis using locally appropriate commuting patterns, and the same windows shall be applied consistently across sessions. During every session, the researchers shall additionally record the actual traffic observations so that the assigned category can be checked against the field condition that actually occurred.

Where vehicle counts are available, the count rate for each session should be reported as a supporting quantitative descriptor. A session may therefore be described as

peak or off-peak based on its predeclared time window, while the observed vehicle count provides an independent check of whether the expected traffic contrast was present. The study shall avoid claiming that a particular vehicle count, by itself, universally defines a structural excitation state for bridges.

For the minimum analysis, two traffic groups are sufficient: peak and off-peak. Weekday-versus-weekend comparisons may be presented as a secondary stratification when enough valid sessions are available. If the collected sessions do not support a reliable multi-group comparison, the thesis shall explicitly report the reduced grouping rather than creating unsupported categories.

Session field	Required record	Purpose	Analysis use
Session ID / date / time	Unique ID, date, start and end time	Traceability	Dataset identification
Traffic condition	Peak or off-peak; weekday/weekend when applicable	Operational classification	Grouping variable
Traffic observation	Vehicle count/count rate when feasible; unusual events	Check actual traffic contrast	Condition validation

Environment	Weather notes; temperature/humidity when available	Contextual documentation	Interpretation/limitation
Sensor state	Mounting location, orientation, integrity check	Measurement consistency	Quality control
Frequency output	Dominant frequency per axis	Primary response variable	Baseline and statistical analysis

Table 2. *Minimum field information recorded for each measurement session*

3.9 Signal Processing and Dominant-Frequency Extraction

Each accepted vibration record will be processed using a consistent signal-processing pipeline. The objective is to obtain one dominant-frequency estimate per analysis axis and per session. The processing sequence will consist of data integrity checking, removal of the sensor *mean* or DC component, optional detrending, application of a predefined window function when required, FFT computation, magnitude-spectrum formation, exclusion of the zero-frequency bin and any frequencies outside the analysis band, and peak selection using a fixed rule.

The frequency-domain conversion is based on the Discrete Fourier Transform computed efficiently through the Fast Fourier Transform. For a sampling frequency f_s and N-point record, the frequency spacing is:

$$\Delta f = \frac{f_s}{N} \tag{1}$$

where Δf is the frequency-bin spacing in hertz, f_s is the sampling frequency in hertz, and **N** is the number of samples in the FFT record. The chosen **N** and f_s shall be selected so that the frequency resolution is appropriate for the bridge frequencies of interest and can be executed within the processing and memory limits of the ESP32-S3.

For a sampled sequence **x[n]**, the DFT magnitude is conceptually obtained from the complex FFT output **X[k]**. The frequency associated with bin **k** is:

$$f_k = k\left(\frac{f_s}{N}\right) \tag{2}$$

where **k** is the FFT-bin index. The dominant frequency f_d for a session and axis is defined as the frequency corresponding to the largest valid spectral peak within the predefined analysis band. The same analysis band and peak-selection criterion must be applied to all sessions to avoid changing the definition of the response variable after observing the results.

Peak picking is computationally lightweight and is suitable for a constrained embedded implementation when the dominant peaks are sufficiently separated. However, because peak picking can be ambiguous for closely spaced modes or weak/noisy responses, the result is treated primarily as a repeatable dominant spectral-frequency indicator. It is not automatically assumed to represent every theoretical natural frequency or a unique structural mode (Hasani & Freddi, 2023).

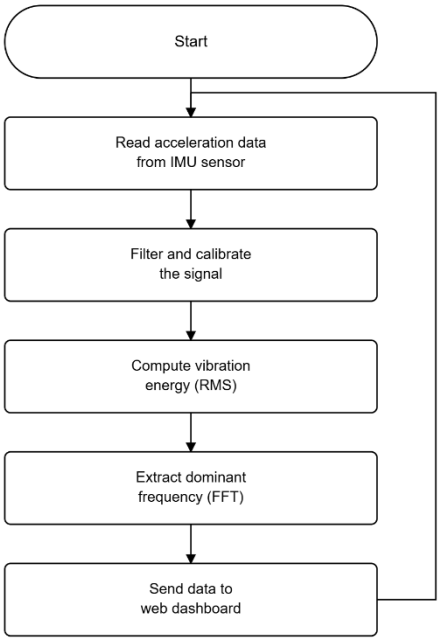


Fig 3. *System Flow Chart*

3.10 Baseline Frequency Construction

The baseline will be constructed from valid field sessions collected under the normal operating conditions represented in the dataset. The baseline is internal to the developed system and the monitored bridge during the study period; it is not a universally accepted structural safety limit and is not externally validated against a separately calibrated reference system.

For each axis, the dominant-frequency observations from the accepted baseline sessions will be summarized using the arithmetic mean and sample standard deviation. The baseline interval is initially represented as:

$$Baseline\ interval = \bar{x} \pm s \quad (3)$$

where $\bar{\mathbf{x}}$ is the mean dominant frequency and **s** is the sample standard deviation. The resulting lower and upper values describe the observed spread of the monitored frequency indicator. The ***coefficient of variation*** is also reported to describe relative dispersion:

$$CV = \left(\frac{S}{\bar{x}}\right) \times 100\% \tag{4}$$

A low **CV** indicates comparatively consistent frequency estimates within the observed operating set, while a larger **CV** indicates greater variation. The final thesis shall report the number of accepted sessions contributing to the baseline, the traffic composition of those sessions, and the axis-specific statistics. Any frequency observation that falls outside the empirical interval shall be flagged for review, but not declared evidence of structural damage by itself.

3.11 Statistical Treatment

Descriptive statistics will first be used to summarize dominant-frequency observations for each axis and traffic condition. These include the number of valid sessions, mean, standard deviation, minimum and maximum, and coefficient of variation. Graphical summaries such as box plots, scatter plots, and mean-with-error-bar plots may be used to show the distribution and variability of the session-level frequency estimates.

The primary inferential comparison is a **one-way ANOVA** when the data structure and assumptions are reasonably satisfied. The factor is traffic condition and the response is the session-level dominant frequency. The null hypothesis is that the mean dominant frequency is equal across the traffic-condition groups. Statistical significance will be evaluated at $\boldsymbol{\alpha = 0.05}$. Where more than two traffic groups are analyzed and the omnibus

ANOVA is significant, an appropriate post-hoc comparison may be used to identify which groups differ, subject to the final sample size and assumptions.

Before ANOVA, the distribution of residuals and the equality of variances will be examined using appropriate diagnostic procedures. Because the study may have a limited number of field sessions, formal normality tests will not be interpreted in isolation; diagnostic plots, group sizes, outliers, and the practical characteristics of the measurements will also be considered. If the assumptions required for ANOVA are not reasonably supported, a Kruskal-Wallis test will be used as the nonparametric alternative for independent traffic groups. The analysis will report the actual sample sizes and the basis for the selected test.

When only two traffic groups are retained, the one-way ANOVA remains mathematically equivalent to the corresponding two-group comparison in terms of the omnibus test. The study will retain the ANOVA framework for consistency with the planned multi-condition methodology.

3.12 Reliability and Repeatability Assessment

Reliability in this study is treated as repeatability of the measurement and processing procedure rather than as calibration accuracy against a certified reference instrument. Repeatability will be examined by holding the sensor location, axis orientation, acquisition settings, session duration, FFT settings, analysis band, and peak-picking rule constant across valid sessions.

The session-level dominant-frequency values will be inspected for excessive spread, obvious processing failures, and systematic drift. The mean, standard deviation, and

coefficient of variation will provide quantitative indicators of repeatability. When a test or pilot session is repeated under substantially similar conditions, the closeness of the resulting dominant-frequency estimates will provide an additional practical check. Any hardware change, firmware revision, sensor remounting, or processing-parameter change that occurs during data collection shall be logged and considered when interpreting the dataset.

3.13 Data Quality Control and Acceptance Criteria

A measurement session will be admitted to the final dataset only when the acquisition record is complete, the sensor remained securely mounted, timestamps and session metadata are present, and the signal is suitable for the predefined FFT procedure. Data showing saturation, severe clipping, obvious discontinuity, corrupted file contents, or a documented hardware malfunction will be excluded with the reason recorded in a data-quality log.

The researchers shall avoid selective removal of observations merely because their frequencies are unusually high or low. Outliers will first be investigated for technical causes. If no technical cause is identified, the observation shall remain in the dataset and be addressed through transparent statistical analysis. This procedure reduces the risk of manipulating the baseline to obtain a desired result.

3.14 Ethical, Safety, and Site Considerations

The study involves field work on an active bridge and therefore requires safety procedures beyond those applicable to laboratory testing. Written authorization from the responsible authority shall be obtained before formal deployment. Researchers shall remain

in designated safe areas, use appropriate personal protective equipment, secure cables and hardware against movement, and avoid any activity that could distract drivers, obstruct traffic, or compromise bridge operations.

The study will not intentionally impose abnormal or unsafe loads on the bridge. Vehicle activity is observed as it occurs during normal operation. No personally identifying information from drivers or pedestrians is required for the vibration-frequency analysis. If photographs or field videos are collected for documentation, they shall be limited to what is necessary for the methodology and shall avoid unnecessary identification of individuals.

3.15 Summary of the Methodological Framework

The methodology links the physical bridge response to the computational output through a controlled and repeatable measurement chain. Traffic-induced vibration is captured by the MEMS IMU, sampled by the ESP32-S3, preprocessed, transformed into the frequency domain using FFT, and reduced to a dominant-frequency value using a fixed peak-picking rule. Repeated session results are then aggregated to describe the bridge's empirical frequency behavior and baseline spread. Traffic condition is explicitly documented so that operational variability can be distinguished from uncontrolled noise. Finally, statistical comparison is used to determine whether the observed dominant-frequency distributions differ across the defined traffic conditions.

This methodology is intentionally limited to frequency characterization and system evaluation. It does not establish a structural damage threshold, diagnose a specific defect, determine load capacity, or replace formal bridge inspection. These boundaries preserve consistency between the methods, the stated objectives, and the scope of the study.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the findings of the study and interprets them in relation to the objectives and literature reviewed. Results should be organized according to the research objectives, with each subsection presenting the findings and immediately discussing their meaning.

4.1 Introduction

Provide a brief overview of the chapter. State that results will be presented per objective and discussed considering related literature.

4.2 Specific Objective 1 (Replace this heading with the descriptive title of your first objective, i.e., “Profile of the Respondents”)

Present the results related to the first objective. Use tables, figures, or graphs where appropriate. Each table, figure, or equation must be referenced in the text, formatted consistently, and accompanied by a clear caption. Immediately follow with discussion that connects the findings to existing studies or theories

4.3 Specific Objective 2 (Replace this heading with the descriptive title of your next objective, i.e., “Relationship Between Variables”)

Present the results related to the second objective. Again, reference any supporting visuals or equations in the text, and provide interpretations that link the findings to the

research problem and reviewed literature. Repeat subsections for additional objectives as needed, always pairing results with discussion.

CHAPTER V

CONCLUSION AND RECOMMENDATION

This chapter summarizes the study, presents conclusions based on the findings, and offers recommendations for practice, policy, or future research.

5.1 Summary

Provide a concise recap of the study's purpose, methodology, and major findings. This should be written in paragraph form, not as bullet points.

5.2 Conclusion

State the conclusions drawn directly from the results. Each conclusion should clearly answer the research objectives. Avoid introducing new data here.

5.3 Recommendations

Offer practical or theoretical recommendations based on the conclusions. These may include suggestions for institutions, practitioners, policymakers, or future researchers. Recommendations should be specific, actionable, and logically derived from the findings.

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APPENDICES

Appendix A. Guide to Appendices

The appendices provide supplementary material that supports the study but is too detailed or lengthy to include in the main text. Each appendix should be labeled clearly, using the format “Appendix A. Title/Description” or “Appendix B. Title/Description”, with the content following immediately after the label and title. Appendices must be referenced in the body of the manuscript where relevant, and they should contain materials such as survey instruments, raw data tables, sample calculations, extended descriptions of procedures, or supplementary figures and diagrams.

Figures, tables, and equations included in the appendices must have numbering that is separate from the main text. To distinguish them, use the appendix letter as a prefix. For example, figures in Appendix A should be labeled as Figure A1, Figure A2, and so on, while tables should be labeled as Table A1, Table A2, and equations as (A1), (A2). In Appendix B, numbering would continue as Figure B1, Table B1, and (B1), respectively.

Appendix B. Typical Appendix Contents

Typical contents for the appendices may include:

1. Copies of survey questionnaires or interview guides
2. Raw data tables or extended datasets
3. Sample calculations or derivations
4. Detailed descriptions of instruments or procedures
5. Additional figures, charts, or diagrams not central to the main discussion

Appendices should be formatted consistently, with clear titles and stand-alone descriptions. They serve as supporting evidence and documentation, ensuring transparency and replicability without interrupting the flow of the main chapters.